

Problem 8.15

A fund accumulates based on the following formula:

$$a(t) = \left(1 + \frac{t}{10}\right) 1.2^t.$$

Find d_3 .

Solution

The effective rate of discount in the 3rd year is

$$d_3 = \frac{a(3) - a(2)}{a(3)}.$$

Substituting the accumulation function,

$$\begin{aligned} d_3 &= \frac{(1 + 0.3)(1.2)^3 - (1 + 0.2)(1.2)^2}{(1 + 0.3)(1.2)^3} \\ &= \frac{(1.3)(1.2)^3 - (1.2)(1.2)^2}{(1.3)(1.2)^3} \\ &= \frac{3}{13} \\ &\approx 0.23076923. \end{aligned}$$

Therefore,

$d_3 = \frac{3}{13} \approx 0.23076923$
